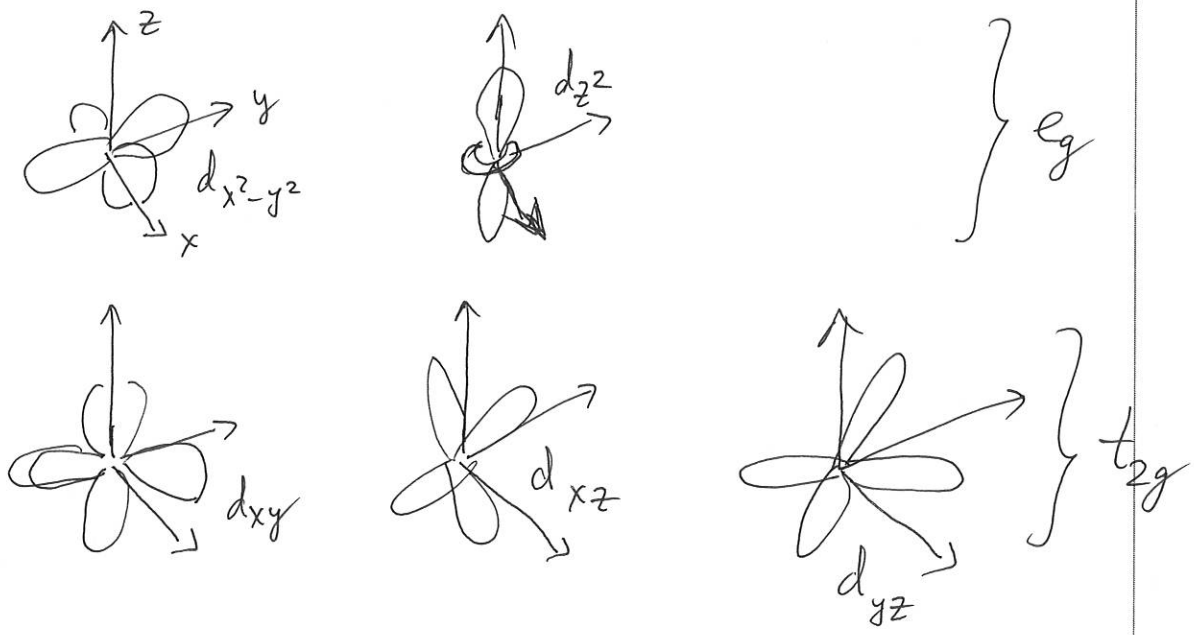
Crystal fields

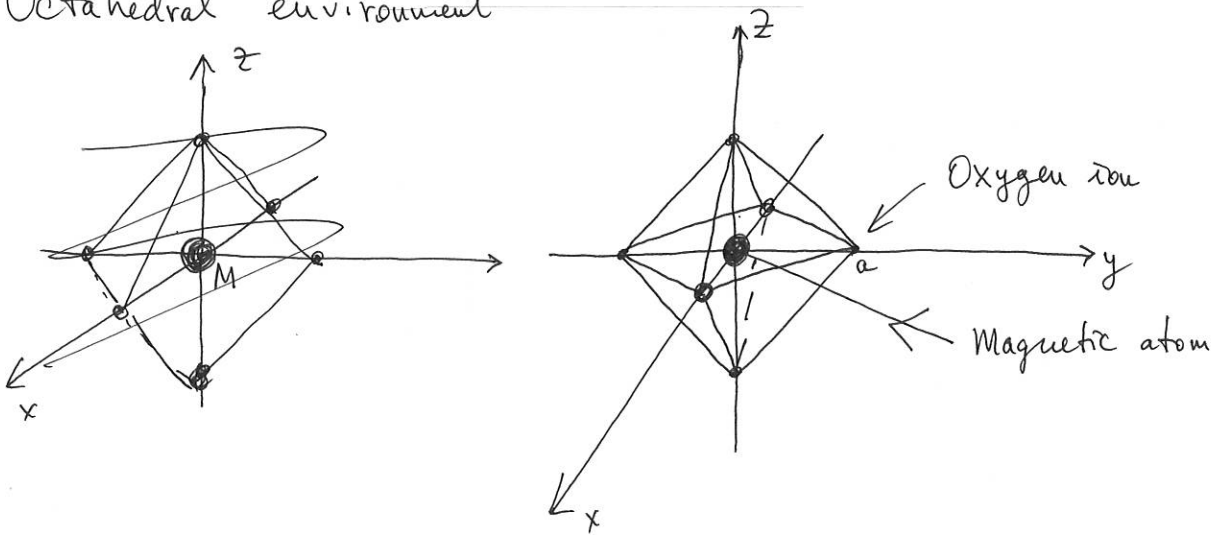
We now describe effect of local environment of magnetic atom (ion) in a crystal.

Wavefunctions of $L \neq 0$ states have rather complicated angular structure, which is sensitive to the local electric potential. This effect for ions of 3d group ($3d^1 - 3d^{10}$) where it competes with Hund's rules (in particular, rule #3) and leads to <u>quenching</u> of orbital momentum

<u>d-orbitals</u>



Octahedral environment



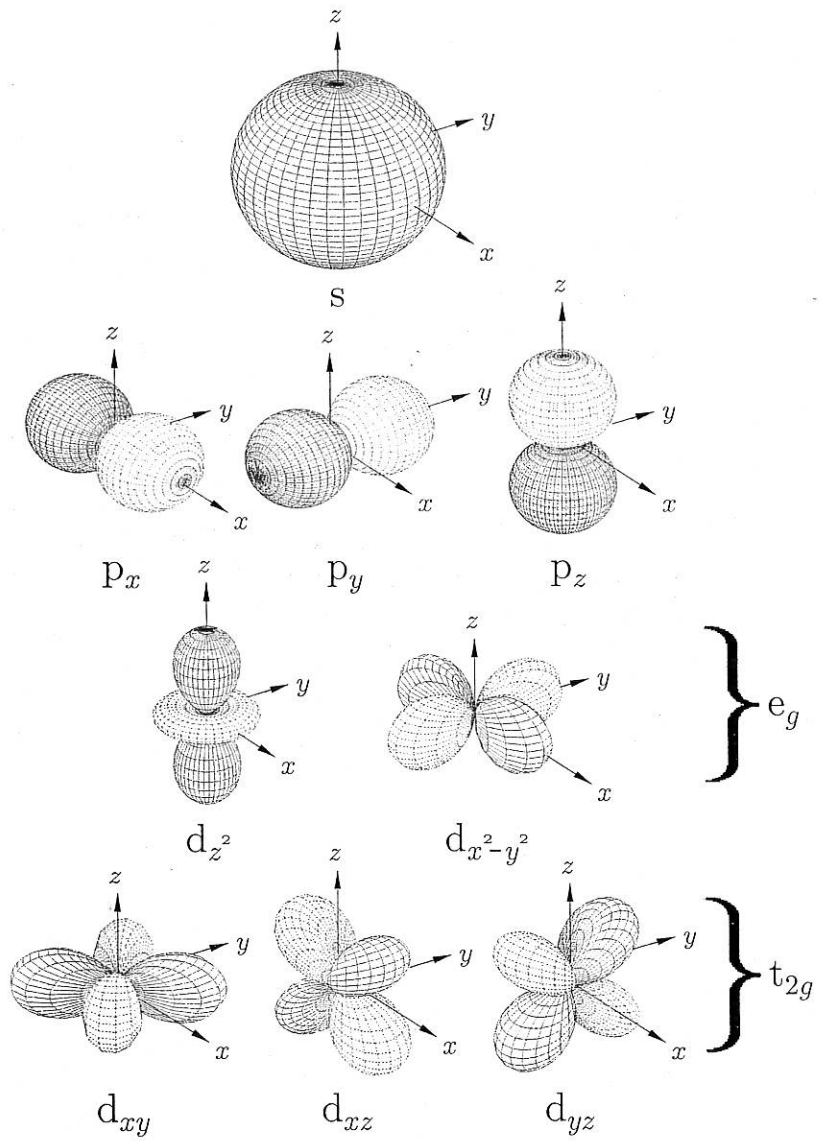


Fig. 3.1 The angular distribution of the s, p and d orbitals. The d_{z^2} and $d_{x^2-y^2}$ levels are grouped together and called the e_g levels. The d_{xy}, d_{xz} and d_{yz} levels are grouped together and called the t_{2g} levels. The d_{z^2} orbital is sometimes referred to as $d_{3z^2-r^2}$.

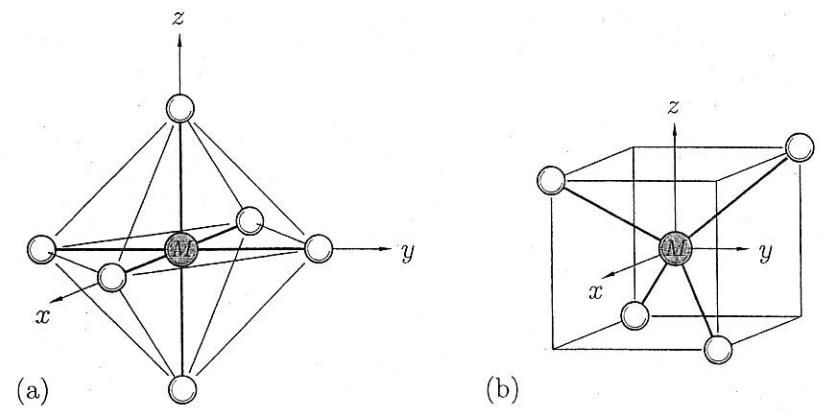


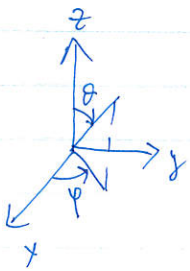
Fig. 3.2 A metal atom M in an (a) octahedral and (b) tetrahedral environment. The octahedral environment is found in many transition metal oxides where oxygen anions sit on the corners of an octahedron with the metal atom at the centre. The tetrahedral environment can be conveniently described by considering alternate corners of a cube as shown.

D orbitals ($l=2$)

are linear combinations of spherical harmonics

$$Y_{xy} = \frac{1}{i\sqrt{2}}\left(Y_2^2 - Y_2^{-2}\right) = \frac{1}{i\sqrt{2}}\left(\frac{15}{32\pi}\right)^{1/2} \sin^2\theta \underbrace{\left(e^{i2\varphi} - e^{-i2\varphi}\right)}_{i\sin 2\varphi} =$$

$$= \left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta \sin 2\varphi = \left(\frac{15}{4\pi}\right)^{1/2} \frac{xy}{r^2}$$



$$Y_{x^2-y^2} = \frac{1}{\sqrt{2}}\left(Y_2^2 + Y_2^{-2}\right) = \left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta \cos 2\varphi = \left(\frac{15}{16\pi}\right)^{1/2} \frac{x^2-y^2}{r^2}$$

$$Y_{yz} = \frac{-1}{i\sqrt{2}}\left(Y_2^1 + Y_2^{-1}\right) = \frac{-1}{i\sqrt{2}}\left(\frac{15}{8\pi}\right)^{1/2} \sin\theta\cos\theta\left(-e^{+i\varphi} + e^{-i\varphi}\right) =$$

$$= \left(\frac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta\sin\varphi = \left(\frac{15}{4\pi}\right)^{1/2} \frac{yz}{r^2}$$

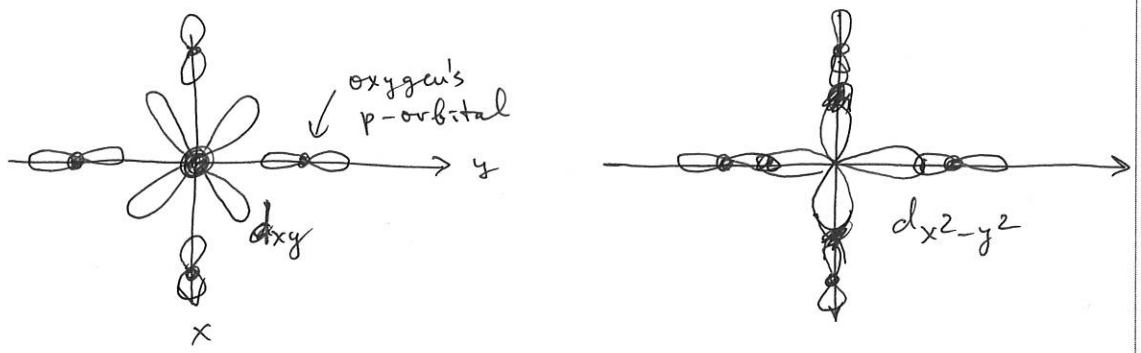
$$Y_{zx} = \frac{1}{\sqrt{2}}\left(-Y_2^1 + Y_2^{-1}\right) = \frac{1}{\sqrt{2}}\left(\frac{15}{8\pi}\right)^{1/2} \sin\theta\cos\theta\left(e^{i\varphi} + e^{-i\varphi}\right)$$

$$= \left(\frac{15}{4\pi}\right)^{1/2} \sin\theta\cos\theta\cos\varphi = \left(\frac{15}{4\pi}\right)^{1/2} \frac{xz}{r^2}$$

$$Y_{z^2} = Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2}\left(3\cos^2\theta - 1\right) = \left(\frac{5}{16\pi}\right)^{1/2} \frac{3z^2 - r^2}{r^2} =$$

$$= \left(\frac{15}{16\pi}\right)^{1/2} \frac{2z^2 - x^2 - y^2}{r^2}$$

Top view explains the basic feature best:



d_{xy} overlaps less that $d_{x^2-y^2}$ $\Rightarrow$ energies of these orbitals must be different!

Consider electric potential of 6 charges in corners of an octahedron : $(\pm a, 0, 0)$, $(0, \pm a, 0)$ and $(0, 0, \pm a)$.

$V(r) = \dfrac{q}{4\pi\varepsilon_0 r}$ where r is measured from the center of octahedron.

$$\Rightarrow V(r) = \frac{q}{4\pi\varepsilon_0}\left\{\frac{1}{\sqrt{(a\pm x)^2+y^2+z^2}} + \frac{1}{\sqrt{x^2+(a\pm y)^2+z^2}} + \frac{1}{\sqrt{x^2+y^2+(a\pm z)^2}}\right\}$$

Start with $\widetilde{V}_x = \dfrac{1}{\sqrt{(a+x)^2+y^2+z^2}} + \dfrac{1}{\sqrt{(a-x)^2+y^2+z^2}} = (x, y, z \ll a) =$

$$= \frac{1}{a}\left\{\frac{1}{\left(1+\frac{2x}{a}+\frac{x^2+y^2+z^2}{a^2}\right)^{1/2}} + \frac{1}{\left(1-\frac{2x}{a}+\frac{x^2+y^2+z^2}{a^2}\right)^{1/2}}\right\} =$$

$$= \frac{1}{a}\left\{\frac{1}{\left(1+\frac{2x}{a}+\frac{r^2}{a^2}\right)^{1/2}} + \frac{1}{\left(1-\frac{2x}{a}+\frac{r^2}{a^2}\right)^{1/2}}\right\}$$

$$V(r) = \frac{q}{4\pi\varepsilon_0 a}\left\{\frac{1}{\sqrt{1+\frac{2x}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1-\frac{2x}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1+\frac{2y}{a}+\frac{r^2}{a^2}}} + \right.$$

$$\left. + \frac{1}{\sqrt{1-\frac{2y}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1+\frac{2z}{a}+\frac{r^2}{a^2}}} + \frac{1}{\sqrt{1-\frac{2z}{a}+\frac{r^2}{a^2}}}\right\}$$

Mathematica

```
In[16]:= Series[1/Sqrt[1 + 2 x + x^2 + y^2 + z^2] + 1/Sqrt[1 - 2 x + x^2 + y^2 + z^2] +
    1/Sqrt[1 + 2 y + x^2 + y^2 + z^2] + 1/Sqrt[1 - 2 y + x^2 + y^2 + z^2] +
    1/Sqrt[1 + 2 z + x^2 + y^2 + z^2] + 1/Sqrt[1 - 2 z + x^2 + y^2 + z^2],
   {x, 0, 4}, {y, 0, 4}, {z, 0, 4}] // FullSimplify[#] & // Normal // Expand
```

$$\text{Out[16]=}\ 6 + \frac{7x^4}{2} - \frac{21x^2y^2}{2} - \cancel{\frac{45x^4y^2}{8}} + \frac{7y^4}{2} - \cancel{\frac{45x^2y^4}{8}} + \cancel{\frac{3465x^4y^4}{32}} - \frac{21x^2z^2}{2} - \cancel{\frac{45x^4z^2}{8}} - \frac{21y^2z^2}{2} + \cancel{\frac{135}{2}x^2y^2z^2} - \cancel{\frac{45y^4z^2}{8}} - \cancel{\frac{20\,475}{32}x^4y^4z^2} + \frac{7z^4}{2} - \cancel{\frac{45x^2z^4}{8}} + \cancel{\frac{3465x^4z^4}{32}} - \cancel{\frac{45y^2z^4}{8}} - \cancel{\frac{20\,475}{32}x^4y^2z^4} + \cancel{\frac{3465y^4z^4}{32}} - \cancel{\frac{20\,475}{32}x^2y^4z^4} + \cancel{\frac{155\,925}{32}x^4y^4z^4} =$$

$$\Rightarrow \tilde{V} = 6 + \frac{7}{2}\left(x^4+y^4+z^4\right) - \frac{21}{2}\left(x^2y^2+x^2z^2+y^2z^2\right) + O(x^6)$$

Observe that $(x^2+y^2+z^2)^2 \equiv (r^2)^2 = x^4+y^4+z^4+2x^2y^2+2x^2z^2+2y^2z^2$

$$\Rightarrow \cancel{x^2+y^2}\ x^2y^2+x^2z^2+y^2z^2 = \frac{1}{2}\left(r^4 - x^4-y^4-z^4\right)$$

$$\tilde{V} = 6 + \frac{7}{2}\left(x^4+y^4+z^4\right) - \frac{21}{2}\,\frac{1}{2}\left(r^4 - (x^4+y^4+z^4)\right) =$$

$$= 6 + \underbrace{\left(\frac{7}{2}+\frac{21}{4}\right)}_{\frac{35}{4}}\left(x^4+y^4+z^4\right) - \frac{21}{4}r^4 =$$

$$= 6 + \frac{35}{4}\left(x^4+y^4+z^4 - \frac{21}{35}r^4\right) = 6 + \frac{35}{4}\left(x^4+y^4+z^4 - \frac{3}{5}r^4\right) + O(r^6).$$

$$\frac{21}{35} = \frac{3}{5}$$

Mathematica shows that

$$V(r) = \frac{q}{4\pi\varepsilon_0 a}\left[6 + \frac{35}{4a^4}\left(x^4+y^4+z^4 - \frac{3}{5}r^4\right) + O\left(\frac{r^6}{a^6}\right)\right]$$

This is crystal field potential produced by 6 anions (oxygens) of charge q each. Electron on magnetic ion M experiences this $V(r)$.

Drop constant and write $V(r) = D\left(x^4+y^4+z^4 - \frac{3}{5}r^4\right)$.

Consider $V(r)$ a perturbation on degenerate d-levels of M-ion ($L=2 \Rightarrow 5$ degenerate levels).

Moreover, $x^4+y^4+z^4 = r^4\left[(\sin\theta\cos\varphi)^4 + (\sin\theta\sin\varphi)^4 + (\cos\theta)^4\right] =$

$$= r^4\left[\sin^4\theta \underbrace{(\cos^4\varphi + \sin^4\varphi)}_{\frac{1}{8}(e^{i4\varphi}+e^{-i4\varphi}+6)} + \cos^4\theta\right] = \frac{r^4}{8}\left[\sin^4\theta\,(6+e^{i4\varphi}+e^{-i4\varphi}) + 8\cos^4\theta\right]$$

$$\Rightarrow V(r) = \frac{Dr^4}{8}\left\{\sin^4\theta\,(6+e^{i4\varphi}+e^{-i4\varphi}) + 8\cos^4\theta - \frac{24}{5}\right\}$$

Wave functions of 3d orbitals: $n=3, l=2, m=0,\pm1,\pm2$

$$\psi_{nlm}(r,\theta,\varphi) = R_{32}(r)\,Y_2^m(\theta,\varphi)$$

$$Y_2^{\pm2} = \sqrt{\frac{15}{32\pi}}\,\sin^2\theta\, e^{\pm i2\varphi}$$

$$Y_2^{\pm1} = \mp\sqrt{\frac{15}{32\pi}\cdot 4}\,\sin\theta\, e^{\pm i\varphi} = \sqrt{\frac{15}{32\pi}}\left(\mp 2\sin\theta\, e^{\pm i\varphi}\right)$$

$$Y_2^0 = \sqrt{\frac{15}{32\pi}\cdot\frac{2}{3}}\,(3\cos^2\theta - 1) = \sqrt{\frac{15}{32\pi}}\left(\sqrt{\frac{2}{3}}\,[3\cos^2\theta - 1]\right)$$

* Absorb normalization factor $\sqrt{\frac{15}{32\pi}}$ into $R_{32}(r)$; $R \to \tilde{R}$

$$\psi_{32m}(r,\theta,\varphi) = \hat{R}_{32}(r) \begin{cases} \sin^2\theta\, e^{\pm i2\varphi} \\ \mp 2\sin\theta\cos\theta\, e^{\pm i\varphi} \\ \sqrt{\frac{2}{3}}\,(3\cos^2\theta - 1) \end{cases} = \begin{cases} |\pm 2\rangle \\ |\pm 1\rangle \\ |0\rangle \end{cases}$$

where $|\pm m\rangle$ denotes L_z eigenstates.

General state $|\psi\rangle = \sum_{j=-2}^{2} a_j |j\rangle \rightarrow \begin{pmatrix} a_2 \\ a_1 \\ a_0 \\ a_{-1} \\ a_{-2} \end{pmatrix}$

Find $\langle\psi|V(r)|\psi\rangle$ in this 5-dimensional basis.

It is easy to see that $\langle i|V|j\rangle = 0$ with $i \neq j$ (<u>for $i \neq \pm 2, j \neq \pm 2$</u>).

Indeed $V = A(e^{i4\varphi} + e^{-i4\varphi}) + B$ while $|i\rangle \sim e^{im_i\varphi}$

$$\Rightarrow \langle i|V|j\rangle \sim \int_0^{2\pi} e^{-im_i\varphi}\left[A(e^{i4\varphi} + e^{-i4\varphi}) + B\right] e^{-im_j\varphi}\, d\varphi$$

For $m_i \neq m_j$ $\int d\varphi\, e^{i(m_j - m_i)\varphi} = 0$ and $\int d\varphi\, e^{i(m_j - m_i \pm 4)\varphi} = 0$

<u>unless</u> $m_j - m_i = \pm 4$ when the exponent $= 0$

$$\Rightarrow \langle i|V|j\rangle \neq 0 \text{ for } \left[\begin{array}{l} i = 2,\ j = -2 \\ i = -2,\ j = 2 \end{array}\right.$$

Clearly $\langle 2|V|-2\rangle = \langle -2|V|2\rangle = V_{2,-2}$

$$V_{2,-2} = \int_0^\infty dr \cdot r^2\, r^4 \left(\frac{D}{8}\right)\tilde{R}_{32}^2 \int_0^\pi d\theta \sin\theta \int d\varphi\, \sin^2\theta\, e^{-i2\varphi} \left\{ \sin^4\theta\,(6 + e^{i4\varphi} + e^{-i4\varphi}) + 8\cos^4\theta - \frac{24}{5} \right\} \sin^2\theta\, e^{-i2\varphi} = \int_0^\infty dr\, r^6\, \tilde{R}_{32}^2 \left(\frac{D}{8}\right) \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \sin^8\theta.$$

Diagonal components $V_{jj} = \langle j|V|j\rangle \neq 0$ and there we can remove $e^{\pm i4\varphi}$ terms from $V(r)$ altogether: $V(r) \rightarrow \frac{Dr^4}{8}\left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$

$$V_{2,2} = V_{-2,-2} = \int dr\, r^6\, \tilde{R}_{32}^2(r)\, \frac{D}{8} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta\, \sin^4\theta \left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$$

$$V_{1,1} = V_{-1,-1} = \int dr\, r^6\, \tilde{R}^2_{32}(r)\, \frac{D}{8} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta\; 4\sin^2\theta \cos^2\theta \left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$$

$$V_{0,0} = \int dr\, r^6\, \tilde{R}^2_{32}(r)\, \frac{D}{8} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \left(\frac{2}{3}\right)\left(3\cos^2\theta - 1\right)^2 \left(6\sin^4\theta + 8\cos^4\theta - \frac{24}{5}\right)$$

Denoting $\alpha \equiv \int_0^{\infty} dr\, r^6\, \tilde{R}^2_{32}(r)\, \frac{D}{8}$, we have (using Mathematica)

$$V_{2,2} = V_{-2,-2} = \alpha\,(2\pi)\,\frac{256}{1575}$$

$$V_{1,1} = V_{-1,-1} = \alpha\,(2\pi)\left(-\frac{1024}{1575}\right)$$

$$V_{0,0} = \alpha\,(2\pi)\,\frac{512}{525} = \alpha\,(2\pi)\,\frac{3\cdot 512}{1575} = \alpha\,(2\pi)\,\frac{1536}{1575}$$

$$V_{2,-2} = \alpha\,(2\pi)\,\frac{256}{315} = \alpha\,(2\pi)\,\frac{5\cdot 256}{1575} = \alpha\,(2\pi)\,\frac{1280}{1575}$$

Denoting now $V_{2,2} = (2\pi)(\alpha)\left(\frac{256}{1575}\right) \equiv A$, we express all other elements as

$$V_{1,1} = -4A \;;\quad V_{0,0} = 6A \;;\quad V_{2,-2} = 5A$$

This remarkable symmetry hints at some bigger picture

$$\Rightarrow \quad V = \begin{pmatrix} A & 0 & 0 & 0 & 5A \\ 0 & -4A & 0 & 0 & 0 \\ 0 & 0 & 6A & 0 & 0 \\ 0 & 0 & 0 & -4A & 0 \\ 5A & 0 & 0 & 0 & A \end{pmatrix} = A \begin{pmatrix} 1 & 0 & 0 & 0 & 5 \\ 0 & -4 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & -4 & 0 \\ 5 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The eigenvalues are: $\lambda_1 = 6A$ (2-fold degenerate)

$\lambda_2 = -4A$ (3-fold)

with eigenvectors:

$$\begin{array}{l} \text{eigenvectors} \\ \text{eigenvalues} \end{array} \left[\begin{array}{ccccc} |0\rangle & \frac{|2\rangle + |-2\rangle}{\sqrt{2}} & |1\rangle & |-1\rangle & \frac{|2\rangle - |-2\rangle}{\sqrt{2}} \\ 6A & 6A & -4A & -4A & -4A \end{array}\right].$$

Suppose $\underline{A<0}$: ground state, energy $-6|A|$, is (tetrahedral environment) 2-fold degenerate $\leftarrow$ $\underline{e_g \text{ states}}$

$$|e_g^1\rangle = |0\rangle$$

$$|e_g^2\rangle = \frac{1}{\sqrt{2}}\left(|2\rangle + |-2\rangle\right)$$

$$\langle e_g^2|\hat{L}^a|e_g^1\rangle = 0 \quad , a = x, y, z \Rightarrow \text{no orbital momentum.}$$

$\underline{A>0}$ (octahedral environment), energy $-4A$,

3-fold degenerate $\underbrace{|1\rangle, |-1\rangle, \frac{1}{\sqrt{2}}\left(|2\rangle - |-2\rangle\right)}_{t_{2g} \text{ states}}$

$$\left.\begin{aligned} t_{2g}^1 &= \frac{1}{\sqrt{2}}\left(|2\rangle - |-2\rangle\right) \sim \sin^2\theta \sin 2\varphi \\ t_{2g}^2 &= \frac{1}{\sqrt{2}}\left(|1\rangle - |-1\rangle\right) \sim \sin\theta \sin\varphi \\ t_{2g}^3 &= \frac{1}{\sqrt{2}}\left(|1\rangle + |-1\rangle\right) \sim \sin\theta \cos\varphi \end{aligned}\right\} L=1 \text{ states}$$

ground state $\vec{L}$ is not given by Hund's rule

[3d elements]

Remaining weak spin-orbit: 2nd order perturbation effect, produces single ion anisotropy $\sim \Lambda_{\alpha\beta}\hat{S}^\alpha\hat{S}^\beta$

In magnetic field, levels are in addition split by

$$\mu_B B m_\ell \Rightarrow \delta V = \mathrm{diag}(2\mu_B B,\ 1\mu_B B,\ 0,\ -\mu_B B,\ -2\mu_B B)$$

$$\Rightarrow V + \delta V = \begin{pmatrix} A+2h & 0 & 0 & 0 & 5A \\ 0 & -4A+h & 0 & 0 & 0 \\ 0 & 0 & 6A & 0 & 0 \\ 0 & 0 & 0 & -4A-h & 0 \\ 5A & 0 & 0 & 0 & A-2h \end{pmatrix}$$

where for brevity $h \equiv \mu_B B$.

Eigenvalues: $\det(V + \delta V - \lambda \mathbb{1}) = 0$

$$0 = \begin{vmatrix} A+2h-\lambda & 0 & 0 & 0 & 5A \\ 0 & -4A+h-\lambda & 0 & 0 & 0 \\ 0 & 0 & 6A-\lambda & 0 & 0 \\ 0 & 0 & 0 & -4A-h-\lambda & 0 \\ 5A & 0 & 0 & 0 & A-2h-\lambda \end{vmatrix} = (A+2h-\lambda)\left[(4A+\lambda)^2 - h^2\right](6A-\lambda) \cdot (A-2h-\lambda) +$$

$$+ 5A \underbrace{\begin{vmatrix} 0 & -4A+h-\lambda & 0 & 0 \\ 0 & 0 & 6A-\lambda & 0 \\ 0 & 0 & 0 & -4A-h-\lambda \\ 5A & 0 & 0 & 0 \end{vmatrix}}_{-5A\left[(4A+\lambda)^2 - h^2\right](6A-\lambda)} = \left[(4A+\lambda)^2 - h^2\right]\left\{\left[(A-\lambda)^2 - 4h^2\right](6A-\lambda) - 25A^2(6A-\lambda)\right\} =$$

$$= (6A-\lambda)\left[(4A+\lambda)^2 - h^2\right]\left\{(A-\lambda)^2 - 4h^2 - 25A^2\right\}$$

$$(A-\lambda)^2 = 4h^2 + 25A^2$$

weak magnetic field

$$\lambda = A \pm \sqrt{25A^2 + 4h^2} \cong A \pm 5A\left(1 + \frac{2h^2}{25A^2}\right) =$$

$$= \begin{cases} 6A + \dfrac{2h^2}{5A} \\ -4A - \dfrac{2h^2}{5A} \end{cases}$$

levels are:

$$\left.\begin{cases} 6A + \dfrac{2\mu_B^2 B^2}{5A} \approx 6A \\ 6A \end{cases}\right\} \text{no splitting, } L=0$$

$$\left.\begin{cases} -4A + \mu_B B \\ -4A - \mu_B B \\ -4A - \dfrac{2\mu_B^2 B^2}{5A} \approx -4A \end{cases}\right\} L=1 \text{ splitting by } B$$

$\Rightarrow$ original $L=2$ is quenched by Crystal field to $\underline{L=1}$ (for $A>0$)

at low $T \ll A$

or to $\underline{\underline{L=0}}$ (for $A<0$)

* Can be easily confirmed by calculating susceptibility from partition function for single ion.

$\chi(T \ll A;\ B \to 0)$.

Note that same susceptibility at $\underline{T \gg A}$ is that of $L=2$ ion! (because than CF quenching does not play a role)

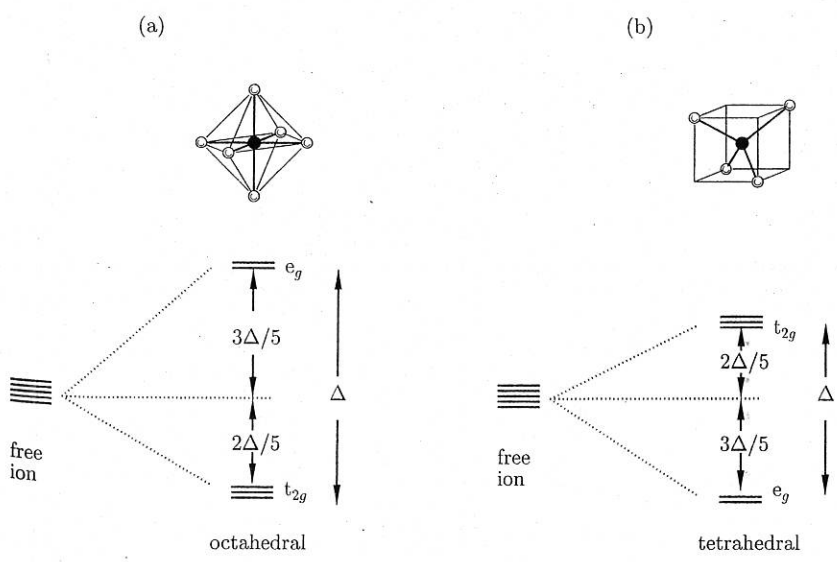


Fig. 3.4 The crystal field in an (a) octahedral and (b) tetrahedral environment.

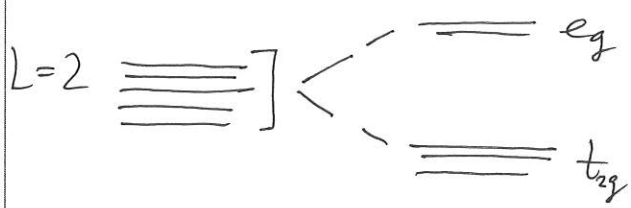
Orbital quenching

→ strong suppression of the angular momentum $\vec{L}$ by the crystal field (CF).
Occurs when CF $\gg$ Spin-Orbit and results in violation of 3rd Hund's rule (that $J = |L-S| \to L+S$)

→ Pronounced in 3d ions (Table 3.1 of Blundell) because their $L=2$ d-orbitals "stick out" far from the center and "feel" CF.

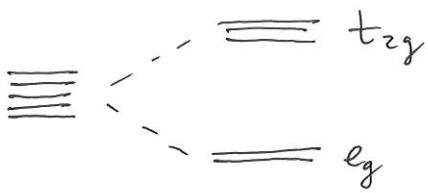
We saw, for <u>octahedral</u> symmetry, that

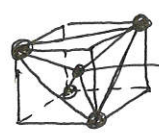
5-fold degenerate



Clearly t_{2g} can be at most $L=1$ state as it is 3-fold degenerate only (d_{xy}, d_{xz}, d_{yz} are related by $\pi/2$ rotations about crystal axes)

for <u>tetrahedral</u>





→ M in the center of tetrahedron

← that would be $L=0$ state because $L \neq$ half-integer: d_{z^2} can not be transformed into $d_{x^2-y^2}$ by discrete rotation consistent with tetrahedron.

More generally, if the ground state is <u>non-degenerate</u>

⇒ w.f. can be choosen real, $|\psi\rangle$.

(e.g. d_{xy}, d_{yz}, d_{xz} $d_{z^2}, d_{x^2-y^2}$ are <u>Real</u>)

$\vec{L}$ is hermitian ⇒ $\langle\psi|\vec{L}|\psi\rangle$ = real.

but $\vec{L} = -i\,\vec{r}\times\vec{\nabla}$ is imaginary

Do note that $\langle\psi|(\vec{L})^2|\psi\rangle \neq 0$!

$$\Rightarrow \langle\psi|\vec{L}|\psi\rangle = \int_V \underbrace{\psi^*}_{\text{real}} \vec{L}\psi\, d\tau = \int \psi \vec{L} \psi = \text{imaginary}$$

⇒ real = imaginary ⇒ $\langle\psi|L|\psi\rangle = 0$ ⇒ orbital momentum is quenched!

Two types of magnetic atoms

$$\psi_{nlm} = R_{nl}(r)\, Y_{lm}(\theta,\varphi) \quad , R_{nl}(r) \sim e^{-\frac{Z}{na_0}r}$$

Transition-metal elements: partially filled <u>3d</u> shell
e.g. cobalt Co

Rare-earth element: partially filled <u>4f</u> shell
e.g. gadolinium

<u>4f</u> electrons are closer to the center than <u>3d</u>

$$\Downarrow$$

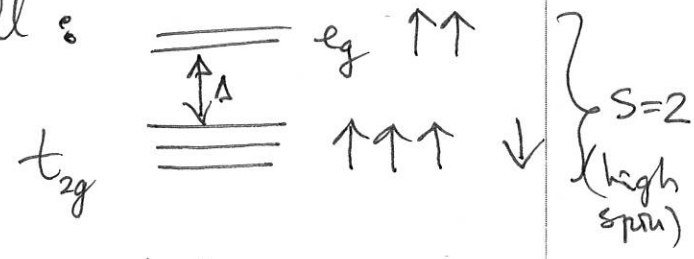
① crystal field is larger in 3d than in 4f

② spin-orbit coupling $\alpha \sim Z^4$: larger in 4f than 3d

	3d	4f
(kelvins) 10^5	e-e Correlation energy!	$\vec{L}, \vec{S}$ according to Hund's rules
Energy 10^4	Crystal field	spin-orbit $\Rightarrow \vec{J} = \vec{L} + \vec{S}$
10^2	Spin-orbit	crystal field
	Orbital moment is reduced (t_{2g}) or quenched (e_{2g})	Ground state multiplet $\vec{J}$ is splitted by crystal field

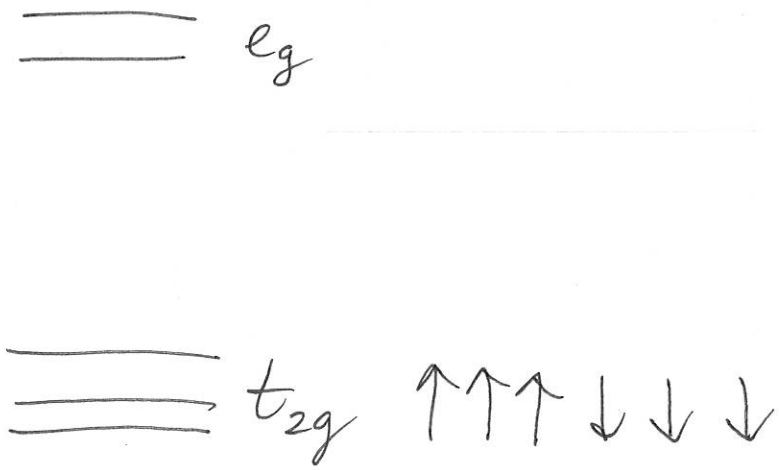
High/low spin states of Fe^{2+} ($3d^6$)

- if crystal field is small:



$\Delta \ll$ e-e correlations so that separation into e_g and t_{2g} is not required

- If large crystal field



S = 0 (low spin)